MIL-W-45481A(MU)
25 FEBRUARY 1963
SUPERSEDING
MIL-W-45481(Ord)
14 MARCH 1960

MILITARY SPECIFICATION

WARHEAD, GUIDED MISSILE, HE, XM9E3 LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers quality assurance provisions, and special requirements not covered by the drawings, pertaining to the loading, assembling and packing for one type of high explosive warhead designated as Warhead, Guided Missile, HE, XM9E3.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

JAN–A–289 — Aluminum Powder, Flaked, Grained and Atomized (for use in Amunition).

MIL–C–401 — Composition B.

MIL–A–2550 — Ammunition and Special Weapons, General Specification for.

MIL–R–11470 — Radiographic Inspection; Qualification of Equipment, Operators and Procedures.

MIL–C–13573 — Calcium Chloride, Anhydrous.

MIL–C–18164 — Composition D–2.

MIL–I–45208 — Inspection Requirements, General Specification for.

MIL–I–45607 — Inspection Equipment, Supply and Maintenance of.

MIL–C–51077 — Calcium Silicate, Technical.

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–109 — Inspection Terms and Definitions.

FSC 1336

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DRAWINGS

ORDNANCE CORPS

8830879 — Crate, Packing, Ammunition, Dual for Warhead, Guided Missile, HE, XM5, XM9 or Warhead, Guided Missile, Inert, XM10

8837892 — Crate, Packing, Ammunition, Single, for Warhead, Guided Missile, HE, XM5, XM9 or Warhead, Guided Missile, Inert, XM10.

8863482 — Warhead, Guided Missile, HE, XM9E3, Loading Assembly.

PUBLICATIONS

ORDNANCE CORPS

ORD–M608–11 — Procedures and Tables for Continuous Sampling by Attributes.

IEL–8863482 — Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Warhead assemblies. The warhead assemblies shall comply with all requirements specified on Drawing (dwg.) 8863482 and with all requirements specified in applicable specifications.

3.3 Composition of explosive charge. The composition of the explosive charge shall be as follows:

Ingredients	Percentage by weight
Composition B (Class 1, MIL-C-401)	73.9 plus or minus 6.
Aluminum (Type C, Class D, JAN-A-289)	20.9 plus or minus 3.
D-2 Desensitizer (MIL-C-18164)	4.7 plus or minus 1.
Calcium Chloride (MIL-C-13573)	0.5 plus or minus 0.1.
Calcium silicate (MIL-C-51077) added	(see below)

The calcium silicate content shall not be less than 1.25 percent, by weight of the actual quantity of TNT in the Composition B in the explosive charge mixture.

3.4 Radiographic examination of the explosive charge. For the determination of the presence of defects in the explosive charge, the charge shall be divided into segments as defined on figure 1.

3.4.1 *Cracks.* Cracks in the explosive charge are permissible provided a charge separation does not exist. A charge separation is defined as the formation of a multisegment charge caused by a crack or cracks, extending entirely through the charge in any direction.

3.4.2 *Cavities.* Cavities in the explosive charge shall not exceed the areas and dimensions specified in table I. Porous areas shall be considered as cavities except that only 80 percent of the projected length and 80 percent of the projected area of a porous section shall be used in computation relative to compliance with table I requirements.

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TABLE I.

	Segment A	Segment B
Sum of projected areas of all cavities (sq. inch)	3/32	3/8
Maximum projected length of any cavity (inch)	1/8	3/8

3.4.3 *Missing explosive components.* No component shall be missing from the explosive charge initiating train.

3.4.4 *Qualification.* Radiographic equipment, operators and procedures shall be qualified in accordance with Specification MIL–R–11470 prior to use in production.

3.5 **Workmanship.** Loading and assembling shall be accomplished in a thorough, workmanlike manner. All parts shall be free of dirt, grease, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Surface coatings shall be continuous except for a few slight scratches not exposing base material. All required marking shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 **General quality assurance provisions.** The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Contractor quality assurance system.* The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specifications MIL–I–45208.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

(b) Number of units of product inspected.

(c) Results obtained for all inspection performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of analysis on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the contractor's written quality assurance pro-

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cedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL–I–45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of warhead assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government shall not necessitate changing the lot interfix number. Each lot shall contain:

(a) Each batch of explosive charge mixture shall contain ingredients from not more than 2 lots, from one manufacturer, of any ingredient.

(b) RDX pellets from not more than one batch or homogeneous blend of batches.

(c) Inert parts assemblies from lots of the same lot interfix number from one manufacturer.

4.2.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL–STD–105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD–M608–11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Pellet, booster (see dwg. 8800281 covering a detail of dwg. 8800264).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Weight	Balance	01001
102.	Diameter	Gage	01002
103.	Thickness	Gage	01003
104.	Pellet cracked or laminated	Visual	01004
Minor: None defined.			

4.2.2.2 Pellet, lower (see dwg. 8800282 covering a detail of dwg. 8863482 and 8861591).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Weight	Balance	02001
102.	Diameter	Gage	02002
103.	Thickness	Gage	02003
104.	Pellet cracked or laminated	Visual	02004
Minor: None defined.			

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4.2.2.3 Pellet cup assembly (see dwg. 8861591 covering a detail of dwg. 8863482).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.25 percent		
101.	Spacer above flush	Visual-Manual	03001
102.	Spacer missing	Visual	03002
Minor: None defined.			

4.2.2.4 Tube assembly (see dwg. 8800283 covering a detail of dwg. 8863482).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.25 percent		
101.	Varnish missing from end	Visual	04001
Minor: None defined.			

4.2.2.5 Parts assembly (see dwg. 8863483 covering a detail of dwg. 8863482).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.65 percent		
101.	Foreign matter on surface inside cone	Visual	05001
102.	Bare spot or scratch in coating inside cone	Visual	05002
103.	Liner missing inside cone	Visual	05003
104.	Insert missing or damaged	Visual	05004
105.	Insert pin missing	Visual	05005
106.	Crack or other damage to assembly	Visual	05006
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	05007

4.2.2.6 Loading assembly, after pouring charge (see dwg. 8863482).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.25 percent		
101.	Cone not filled and leveled	Visual	06001
Minor: None defined.			

4.2.2.7 Loading assembly, prior to inert loading (see dwg. 8863482).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Tube above flush or excessively below flush	Gage	07001
102.	Booster incorrectly assembled or loose	Visual-Manual	07002
103.	Booster cracked or split	Visual	07003
104.	Cap on cone missing	Visual	07004
105.	Explosive on cone exterior	Visual	07005
106.	Pellet cavity not plugged securely	Manual	07006

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Categories	Defects	Method of Inspection	Code No.
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	07007

4.2.2.8 Loading assembly, prior to closure (see dwg. 8863482).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 0.40 percent		
201.	Cavity not filled	Visual	08001
202.	Funnel marks not leveled	Visual	08002
203.	Pad missing	Visual	08003

4.2.2.9 Loading assembly (see dwg. 8863482).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Pellet cup above flush in S&A cavity	Gage	09001
102.	Closing plug bond inadequate	Manual	09002
103.	Flange cracked or damaged	Visual	09003
104.	Paint in threads or S&A cavity	Visual	09004
105.	Evidence of separation or peeling of overlaps of outer body	Visual	09005
Minor:	AQL 1.50 percent		
201.	Exterior coating damaged (see 3.5)	Visual	09006
202.	Marking missing, misleading or unidentifiable	Visual	09007
203.	Evidence of poor workmanship (see 3.5)	Visual	09008

4.2.2.10 Crate, prior to sealing (see dwg. 8830879 or 8837892).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.25 percent		
101.	Warhead loose or improperly assembled	Visual-Manual	10001

4.2.2.11 Crate, sealed (see dwg. 8830879 or 8837892).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Board broken or split	Visual	11001
102.	Skid missing or damaged	Visual	11002
103.	Contents exposed	Visual	11003
104.	Lag screw missing or loose	Visual-Manual	11004
105.	Strapping missing, broken or loose	Visual-Manual	11005

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Categories	Defects	Method of Inspection	Code No.
Minor:	**AQL 1.00 percent**		
201.	Marking missing, misleading or unidentifiable	Visual	11006
202.	Strapping improperly assembled	Visual	11007

4.2.3 *Testing.*

4.2.3.1 *Composition of explosive charge (see 3.3) — Major defect — Code No. 12001.* Three sample wafers of approximately 100 grams (gm.) each shall be taken from the charge at the loading spout. They shall be taken near the beginning, at approximately the middle and near the end of the batch. (If the pour is part of an XM5 Warhead batch, the determination need not be repeated.) One quarter of each wafer shall be ground, thoroughly blended and a portion selected for analysis. Failure of the portion in meeting one or more of the composition requirements shall reject that part of the batch represented by the sample.

4.2.3.2 *Radiographic examination of explosive charge (see 3.4) — Major defect — Code No. 13001.* This examination shall be performed 100 percent using the equipment, operators and procedures qualified under 3.4.4 as specified in 4.3.2. Any assembly failing to meet one or more of the requirements shall be classed defective and removed from the lot.

4.2.3.3 *Assembly weight (see dwg. 8863482) — Major defect — Code No. 09009.* This determination shall be performed 100 percent. Any assembly failing to meet the requirement shall be classed defective and removed from the lot.

4.2.4 *Inspection equipment.* Index of Inspection Equipment Number IEL–8863482 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL–I–45607.

4.3 Test methods and procedures.

4.3.1 *Composition of explosive charge.* (See 6.4)

4.3.1.1 *Calcium silicate and calcium chloride content.* The government inspector shall verify that the correct weights of calcium silicate and calcium chloride are added to each batch.

4.3.1.2 *Explosive content determination.* An accurately weighed 1.0 gram portion of the sample (see 4.2.3.1) shall be transferred to a tared medium porosity sintered glass crucible. The sample shall be extracted 8 times with 20 milliliter (ml.) portions of cold acetone (approximately 5° Centigrade (C.)). After completion of the extraction, the crucible and residue shall be dried in an oven maintained at approximately 60° C. for 5 minutes, cooled in a desiccator and weighed. (Reserve the crucible and content for the D–2 determination.) The percent explosive shall be calculated as follows:

$$*\text{percent explosive} = \frac{100(W2-W3)}{WS}$$

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where:

W2 = weight of crucible and sample.

W3 = weight of crucible and residue after drying.

WS = original weight of sample.

*This percentage includes the nitrocellulose from the D-2 desensitizer.

4.3.1.3 *D–2 desensitizer content determination.* The residue in the crucible reserved from the explosive content determination shall be extracted with four 10 ml. portions of benzene and then rinsed with two 10 ml. portions of chloroform. The crucible and contents shall then be dried in an oven at approximately 65° C. for a half hour, cooled in a desiccator and weighed. The precent D–2 shall be calculated as follows:

$$\text{percent D–2} = \frac{116(W3–W4)}{WS}$$

where:

W3 = weight of crucible and residue from the explosive determination.

W4 = weight of crucible and residue after drying.

WS = original weight of the sample.

116 = factor for nitrocellulose removed in explosive determination.

4.3.1.4 *Aluminum content determination.* The aluminum content shall be calculated as the difference between the sum of the analytical results of 4.3.1.1 and 4.3.1.2 and 100.

4.3.2 *Radiographic examination.* The warhead shall be radiographically examined by directing the beam perpendicular to the plane of the warhead depicted in figure I and after a 90° rotation on its longitudinal axis from that position. Examination of the negatives shall be made for the absence of an explosive train component, for charge defects in excess of those permitted by 3.4.2 and for a crack, or cracks, causing a charge separation. Any unit exhibiting any of the defects shall be classed defective.

5. PREPARATION OF DELIVERY

5.1 Packing and marking.

5.1.1 *Level A.* The loaded warhead shall be packed and marked in accordance with dwg. 8830879 or 8837892, as applicable.

5.2 Data cards. Data card information shall be prepared for each lot in accordance with the procedures specified in Specification MIL–A–2550.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.

(b) That single or dual shipping crates, or both, will be required.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced inspection equipment lists in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists (IEL) indicate the availability of inspection equipment design by showing in the "number" column of the list of inspection equipment (00 Form 1242–3) the numbers of drawings of existing equipment designs or codes indicating "commercial item" or "contractor design." Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.2 and 6.3.3. Action required by the contractor

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with respect to "commercial item" inspection equipment is described in paragraph 6.3.4. The contractor will be required to prepare drawings for all the equipment coded as "contractor design" in the number column. The code will further indicate whether detailed or specification control drawings are required. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approoval as specified in 6.3.7.

6.3.2 *Ordnance designs.* Ordnance designs are detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and nonmandatory. When an IEL references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with nonmandatory designs and with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor prepared designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.3 *Specification control drawings.* Specification control drawings depict the device in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance of other characteristics. Contractors must prepare drawings of their designs in support of specification control drawings. The IEL code will indicate whether drawings must be detailed. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirement of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Commercial items.* Commercial item inspection equipment is equipment that has a universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial item inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.5 *IEL codes.* The inspection equipment referenced in paragraphs 6.3.1, 6.3.2, 6.3.3, and 6.3.4 will be designated in the IEL by the following codes:

CDOF	— Contractor's design responsibility on Ordnance format; ORDM–608–12.
CDCF	— Contractor design responsibility on contractor format.
OD	— Ordnance design.
ODMU	— Ordnance design, mandatory for use.
CE	— Commercial equipment.
SCD	— Specification control drawing.

6.3.6 *Use of standard measuring equipment.* If the contractor desires to use standard measuring equipment in lieu of government or contractor designed gages, diagrammatic sketches or other descriptions of the use of such equipment shall be submitted for approval as specified in 6.3.7.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. Design review will nor-

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mally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Composition of explosive charge. The contractor may use, with prior approval of the responsible Government technical agency, other methods and procedures than those detailed in 4.3.1 for determining the analysis of the explosive charge.

6.5 Inspection log. The Government inspector will maintain a log in which will be recorded the dates and times of his surveillance operations, the characteristics, tests or processes surveyed, deficiencies and improvements noted and the action taken by him, or the contractor, as a result of his surveillance.

6.6 Reusable shipping containers. The contractor should, when possible, utilize for shipment of the loaded units the containers in which the unloaded warheads are received.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government hereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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Preparing activity:
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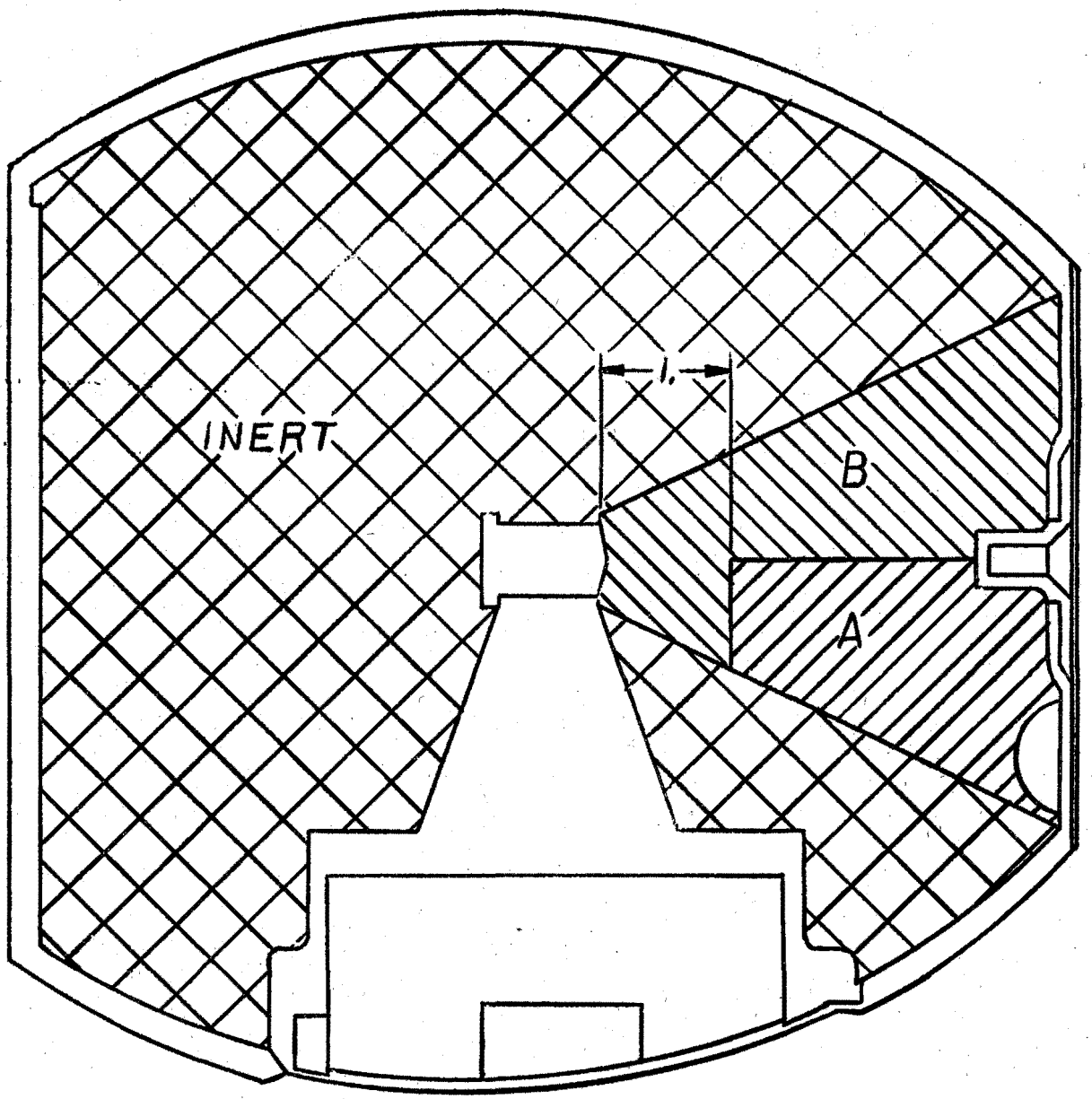


FIGURE 1. *XM9E3 warhead cavitation zones.*

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